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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Center of Mass and Collision

JEE Main Physics · Previous Year Questions with Answer Key

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Numerical	29 questions
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Answer key	at the end

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Numerical 29 questions

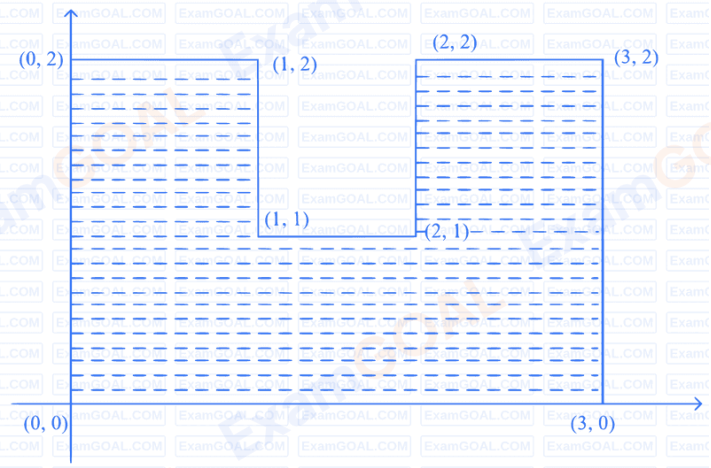
Q1. JEE Main 2026 (Online) 8th April Evening Shift

Two masses of 3.4 kg and 2.5 kg are accelerated from an initial speed of $5\ \mathrm{m/s}$ and $12\ \mathrm{m/s}$, respectively. The distances traversed by the masses in the 5th second are 104 m and 129 m , respectively. The ratio of their momenta after 10 s is $\frac{x}{8}$. The value of x is ____。

Open solution

Q2. JEE Main 2024 (Online) 8th April Morning Shift

A uniform thin metal plate of mass $10\ \mathrm{kg}$ with dimensions is shown. The ratio of x and y coordinates of center of mass of plate is $\frac{n}{9}$. The value of n is _____.



Open solution

Q3. JEE Main 2024 (Online) 4th April Evening Shift

In a system two particles of masses $m_1 = 3\ \mathrm{kg}$ and $m_2 = 2\ \mathrm{kg}$ are placed at certain distance from each other. The particle of mass m_1 is moved towards the center of mass of the system through a distance $2\ \mathrm{cm}$. In order to keep the center of mass of the system at the original position, the particle of mass m_2 should move towards the center of mass by the distance _____ cm.

Open solution

Q4. JEE Main 2024 (Online) 1st February Morning Shift

Three identical spheres each of mass $2\mathrm{M}$ are placed at the corners of a right angled triangle with mutually perpendicular sides equal to $4\ \mathrm{m}$ each. Taking point of intersection of these two sides as origin, the magnitude of position vector of the centre of mass of the system is $\frac{4\sqrt{2}}{x}$, where the value of x is _____ .

Open solution

Q5. JEE Main 2024 (Online) 31st January Morning Shift

A body starts falling freely from height H hits an inclined plane in its path at height h. As a result of this perfectly elastic impact, the direction of the velocity of the body becomes horizontal. The value of $\frac{H}{h}$ for which the body will take the maximum time to reach the ground is _____.

Open solution

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Q6. JEE Main 2023 (Online) 8th April Morning Shift

The momentum of a body is increased by 50%. The percentage increase in the kinetic energy of the body is _____ $\%$.

Open solution

Q7. JEE Main 2023 (Online) 31st January Evening Shift

A ball is dropped from a height of $20\ \mathrm{m}$. If the coefficient of restitution for the collision between ball and floor is 0.5, after hitting the floor, the ball rebounds to a height of _____ m.

Open solution

Q8. JEE Main 2023 (Online) 25th January Evening Shift

A body of mass 1 kg collides head on elastically with a stationary body of mass 3 kg. After collision, the smaller body reverses its direction of motion and moves with a speed of 2 m/s. The initial speed of the smaller body before collision is _____ ms^{-1}.

Open solution

Q9. JEE Main 2022 (Online) 30th July Evening Shift

A body moving horizontally on a smooth surface with a speed of $20\ \mathrm{m\,s}^{-1}$ splits into two parts and continue to move in the same direction . The masses of two parts are in the ratio of $1:2$. The smaller part moves with a speed of $30\ \mathrm{m\,s}^{-1}$. The fractional change is the kinetic energy is given by $\frac{1}{x}$. The value of x is_____

Open solution

Q10. JEE Main 2022 (Online) 28th July Evening Shift

The distance of centre of mass from end A of a one dimensional rod (AB) having mass density $\rho = \rho_0\left(1-\frac{x^2}{L^2}\right)\mathrm{kg/m}$ and length L (in meter) is $\frac{3L}{\alpha}\mathrm{m}$. The value of α is _____. (where x is the distance from end A)

Open solution

Q11. JEE Main 2022 (Online) 25th July Evening Shift

Three identical spheres each of mass M are placed at the corners of a right angled triangle with mutually perpendicular sides equal to 3 m each. Taking point of intersection of mutually perpendicular sides as origin, the magnitude of position vector of centre of mass of the system will be $\sqrt{x}$ m. The value of x is _____.

Open solution

Q12. JEE Main 2022 (Online) 28th June Morning Shift

A man of 60 kg is running on the road and suddenly jumps into a stationary trolly car of mass 120 kg. Then, the trolly car starts moving with velocity 2 ms^{-1}. The velocity of the running man was _____ ms^{-1}, when he jumps into the car.

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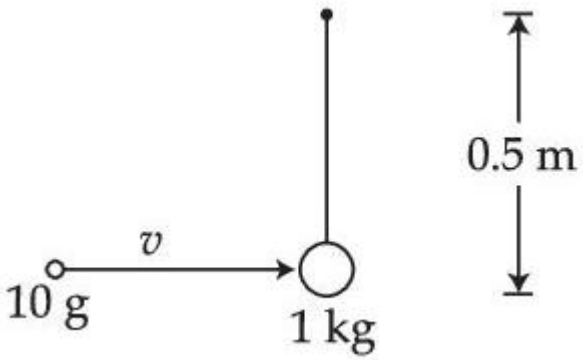
Q13. JEE Main 2022 (Online) 26th June Evening Shift

A batsman hits back a ball of mass 0.4 kg straight in the direction of the bowler without changing its initial speed of 15 ms^{-1}. The impulse imparted to the ball is _____ Ns.

Open solution

Q14. JEE Main 2021 (Online) 27th August Evening Shift

A bullet of 10 g, moving with velocity v, collides head-on with the stationary bob of a pendulum and recoils with velocity 100 m/s. The length of the pendulum is 0.5 m and mass of the bob is 1 kg. The minimum value of v = _____ m/s so that the pendulum describes a circle. (Assume the string to be inextensible and g = 10 m/s^2)



Open solution

Q15. JEE Main 2021 (Online) 25th July Morning Shift

A body of mass 2 kg moving with a speed of 4 m/s makes an elastic collision with another body at rest and continues to move in the original direction but with one fourth of its initial speed. The speed of the two body centre of mass is $\frac{x}{10}$ m/s. Then the value of x is _____.

Open solution

Q16. JEE Main 2021 (Online) 22th July Evening Shift

The position of the centre of mass of a uniform semi-circular wire of radius 'R' placed in x-y plane with its centre at the origin and the line joining its ends as x-axis is given by $\left(0, \frac{xR}{\pi}\right)$. Then, the value of | x | is _____.

Open solution

Q17. JEE Main 2021 (Online) 20th July Morning Shift

A rod of mass M and length L is lying on a horizontal frictionless surface. A particle of mass 'm' travelling along the surface hits at one end of the rod with a velocity 'u' in a direction perpendicular to the rod. The collision is completely elastic. After collision, particle comes to rest. The ratio of masses $\left(\frac{m}{M}\right)$ is $\frac{1}{x}$. The value of 'x' will be _____.

Open solution

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Q18. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

The projectile motion of a particle of mass 5 g is shown in the figure.

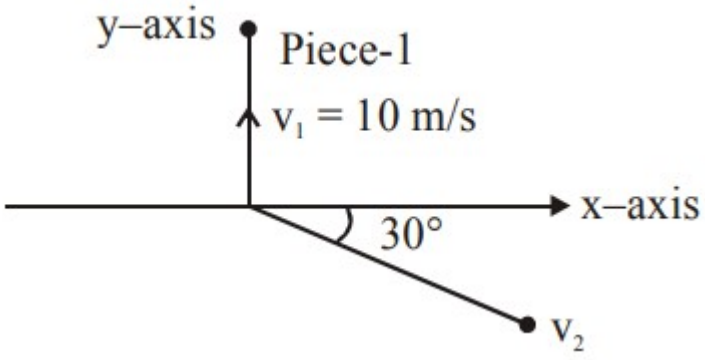


The initial velocity of the particle is $5\sqrt{2}$ ms^{-1} and the air resistance is assumed to be negligible. The magnitude of the change in momentum between the points A and B is $x \times 10^{-2}$ $kgms^{-1}$. The value of x, to the nearest integer, is _____.

Q19. JEE Main 2021 (Online) 18th March Morning Shift

Open solution

A ball of mass 10 kg moving with a velocity $10\sqrt{3}$ m/s along the x-axis, hits another ball of mass 20 kg which is at rest. After the collision, first ball comes to rest while the second ball disintegrates into two equal pieces. One piece starts moving along y-axis with a speed of 10 m/s. The second piece starts moving at an angle of 30° with respect to the x-axis. The velocity of the ball moving at 30° with x-axis is x m/s. The configuration of pieces after collision is shown in the figure below. The value of x to the nearest integer is _____.



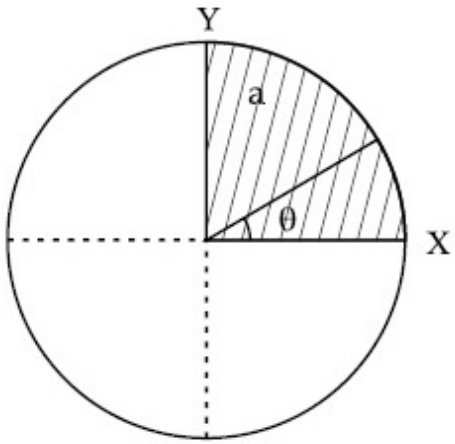
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Q20. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

The disc of mass M with uniform surface mass density σ is shown in the figure. The centre of mass of the quarter disc (the shaded area) is at the position $\frac{x}{3}\frac{a}{\pi}, \frac{x}{3}\frac{a}{\pi}$ where x is _____. (Round off to the Nearest Integer).

[a is an area as shown in the figure]



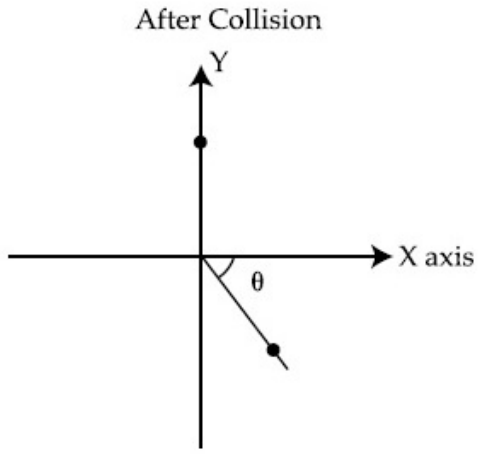
Q21. JEE Main 2021 (Online) 16th March Morning Shift

Open solution

A ball of mass 10 kg moving with a velocity $10\sqrt{3}$ m s^{-1} along X-axis, hits another ball of mass 20 kg which is at rest. After collision, the first ball comes to rest and the second one disintegrates into two equal pieces. One of the pieces starts moving along Y-axis at a speed of 10 m/s. The second piece starts moving at a speed of 20 m/s at an angle θ (degree) with respect to the X-axis.

The configuration of pieces after collision is shown in the figure.

The value of θ to the nearest integer is _____.



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Q22. JEE Main 2021 (Online) 25th February Evening Shift

Two particles having masses 4 g and 16 g respectively are moving with equal kinetic energies. The ratio of the magnitudes of their linear momentum is n : 2. The value of n will be _____.

Open solution

Q23. JEE Main 2021 (Online) 24th February Evening Shift

Two solids A and B of mass 1 kg and 2 kg respectively are moving with equal linear momentum. The ratio of their kinetic energies $(K.E.)_A : (K.E.)_B$ will be $\frac{A}{1}$, so the value of A will be _____.

Open solution

Q24. JEE Main 2021 (Online) 24th February Morning Shift

A ball with a speed of 9 m/s collides with another identical ball at rest. After the collision, the direction of each ball makes an angle of 30° with the original direction. The ratio of velocities of the balls after collision is x : y, where x is _____.

Open solution

Q25. JEE Main 2020 (Online) 6th September Evening Slot

The centre of mass of solid hemisphere of radius 8 cm is x from the centre of the flat surface. Then value of x is _____.

Open solution

Q26. JEE Main 2020 (Online) 6th September Morning Slot

Two bodies of the same mass are moving with the same speed, but in different directions in a plane. They have a completely inelastic collision and move together thereafter with a final speed which is half of their initial speed. The angle between the initial velocities of the two bodies (in degree) is _____.

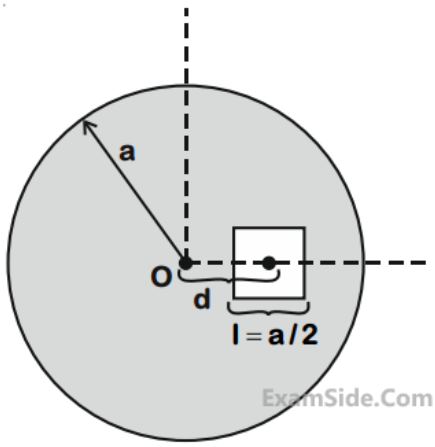
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Q27. JEE Main 2020 (Online) 2nd September Evening Slot

A square shaped hole of side l = $\frac{a}{2}$ is carved out at a distance d = $\frac{a}{2}$ from the centre 'O' of a uniform circular disk of radius a. If the distance of the centre of mass of the remaining portion from O is $-\frac{a}{X}$, value of X (to the nearest integer) is :

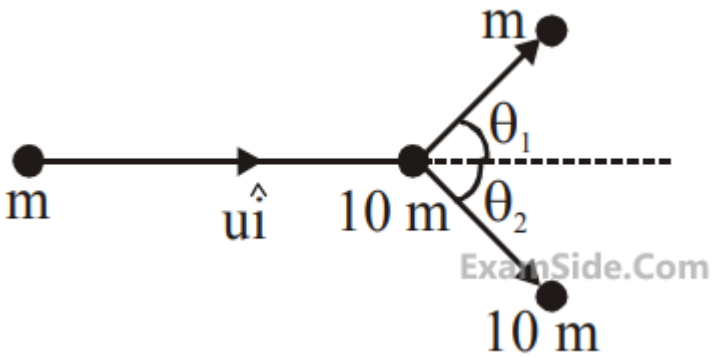
Open solution



Q28. JEE Main 2020 (Online) 2nd September Evening Slot

A particle of mass m is moving along the x-axis with initial velocity $u\hat{i}$. It collides elastically with a particle of mass 10 m at rest and then moves with half its initial kinetic energy (see figure). If $\sin\theta_1 = \sqrt{n}\sin\theta_2$ then value of n is _____.

Open solution



Q29. JEE Main 2020 (Online) 8th January Morning Slot

A body A, of mass m = 0.1 kg has an initial velocity of $3\hat{i}$ ms^{-1} . It collides elastically with another body, B of the same mass which has an initial velocity of $5\hat{j}$ ms^{-1}. After collision, A moves with a velocity $\vec{v} = 4\left(\hat{i}+\hat{j}\right)$. The energy of B after collision is written as $\frac{x}{10}$. The value of x is _____.

Open solution

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MCQ (Single Correct Answer) 117 questions

Q30. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

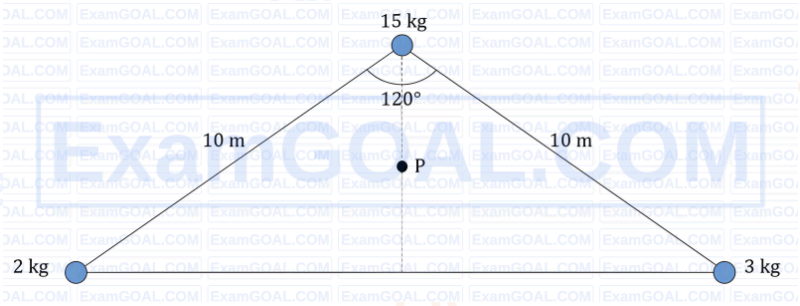
Two identical bodies A and B of equal masses have initial velocities $\overrightarrow{v_1} = 4\hat{i}\,\text{m/s}$ and $\overrightarrow{v_2} = 4\hat{j}\,\text{m/s}$ respectively. The body A has acceleration $\overrightarrow{a_1} = 6\hat{i} + 6\hat{j}\,\text{m/s}^2$ while the acceleration of the other body B is zero. The centre of mass of the two bodies moves in ____ path.

(A) circular (B) parabolic (C) straight line (D) elliptical

Q31. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

The position of center of mass of three masses $2\,\text{kg}, 3\,\text{kg}$ and 15 kg placed with respect to mid point (p) of normal bisector, as shown in the figure is ____ .



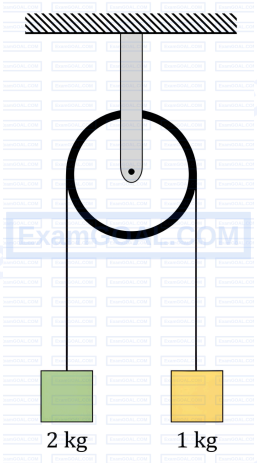
(A) $\left(\frac{\sqrt{3}}{4}, 1.25\right)$ (B) $\left(\frac{\sqrt{3}}{4}, 1.0\right)$

(C) $(0, 0)$ (D) $(1.25, 0)$

Q32. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

Two blocks of masses 2 kg and 1 kg respectively, are tied to the ends of a string which passes over a light frictionless pulley as shown in the figure below. The masses are held at rest at the same horizontal level and then released. The distance traversed by the centre of mass in 2 s is _____ m. (Take $g = 10\ m/s^2$)



(A) 3.33 (B) 3.12 (C) 2.22 (D) 1.42

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Q33. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

A body of mass 14 kg initially at rest explodes and breaks into three fragments of masses in the ratio $2:2:3$. The two pieces of equal masses fly off perpendicular to each other with a speed of $18\ \mathrm{m/s}$ each. The velocity of the heavier fragment is

____ $\mathrm{m/s}$.

(A) $24\sqrt{2}$ (B) 12 (C) $12\sqrt{2}$ (D) $10\sqrt{2}$

Q34. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

In a perfectly inelastic collision, two spheres made of the same material with masses 15 kg and 25 kg , moving in opposite directions with speeds of $10\ \mathrm{m/s}$ and $30\ \mathrm{m/s}$, respectively, strike each other and stick together. The rise in temperature (in $^\circ\mathrm{C}$), if all the heat produced during the collision is retained by these spheres, is :

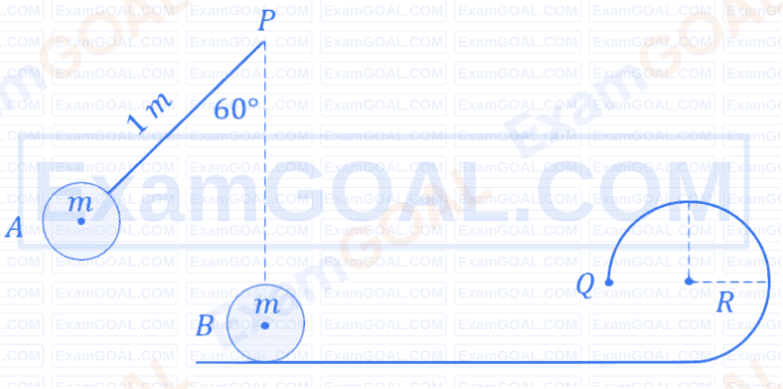
(specific heat of sphere material $31\mathrm{cal/kg.}^\circ\mathrm{C}$ and $1\mathrm{cal} = 4.2\ \mathrm{J}$)

(A) 1.75 (B) 1.44 (C) 1.95 (D) 1.15

Q35. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

A small bob A of mass m is attached to a massless rigid rod of length 1 m pivoted at point P and kept at an angle of 60° with vertical as shown in figure. At distance of 1 m below point P, an identical bob B is kept at rest on a smooth horizontal surface that extends to a circular track of radius R as shown in figure. If bob B just manages to complete the circular path of radius R upto a point Q after being hit elastically by bob A, then radius R is ____ m.



(A) $\frac{1}{5}$ (B) $\frac{2-\sqrt{3}}{5}$ (C) $\frac{3}{5}$ (D) $\frac{2+\sqrt{3}}{5}$

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Q36. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Given below are two statements :

Statement I : For a mechanical system of many particles total kinetic energy is the sum of kinetic energies of all the particles.

Statement II : The total kinetic energy can be the sum of kinetic energy of the center of mass w.r.t. the origin and the kinetic energy of all the particles w.r.t. the center of mass as the reference.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

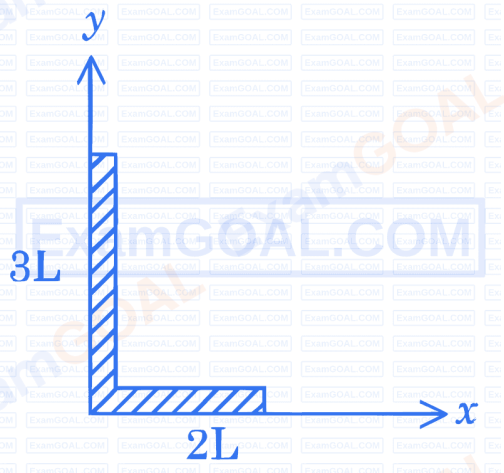
(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

Q37. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

A rod of length 5 L is bent right angle keeping one side length as 2 L .



The position of the centre of mass of the system :

(Consider $L = 10\ cm$)

(A) $4\hat{i} + 9\hat{j}$ (B) $2\hat{i} + 3\hat{j}$ (C) $5\hat{i} + 8\hat{j}$ (D) $3\hat{i} + 7\hat{j}$

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Q38. JEE Main 2025 (Online) 3rd April Evening Shift

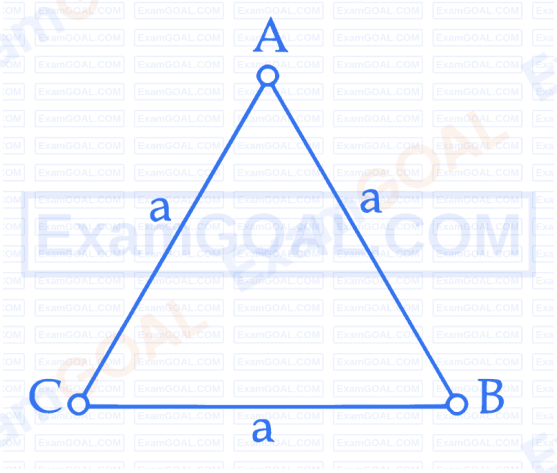


Open solution

Consider two blocks A and B of masses $m_1 = 10\,\text{kg}$ and $\text{m}_2 = 5\,\text{kg}$ that are placed on a frictionless table. The block A moves with a constant speed $v = 3\,\text{m/s}$ towards the block B kept at rest. A spring with spring constant $\text{k} = 3000\,\text{N/m}$ is attached with the block B as shown in the figure. After the collision, suppose that the blocks A and B, along with the spring in constant compression state, move together, then the compression in the spring is, (Neglect the mass of the spring)

(A) 0.3 m (B) 0.1 m (C) 0.4 m (D) 0.2 m

Q39. JEE Main 2025 (Online) 29th January Evening Shift



Open solution

Three equal masses m are kept at vertices (A, B, C) of an equilateral triangle of side a in free space. At $t = 0$, they are given an initial velocity $\overrightarrow{V_A} = V_0\overrightarrow{AC}, \overrightarrow{V_B} = V_0\overrightarrow{BA}$ and $\overrightarrow{V_C} = V_0\overrightarrow{CB}$. Here, $\overrightarrow{AC}, \overrightarrow{CB}$ and $\overrightarrow{BA}$ are unit vectors along the edges of the triangle. If the three masses interact gravitationally, then the magnitude of the net angular momentum of the system at the point of collision is :

(A) $\frac{1}{2}$ a mV_0 (B) 3 a mV_0
(C) $\frac{3}{2}$ a mV_0 (D) $\frac{\sqrt{3}}{2}$ a mV_0

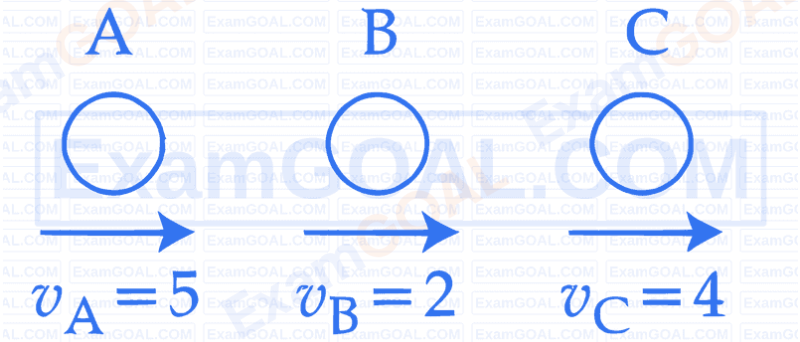
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Q40. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A):



Three identical spheres of same mass undergo one dimensional motion as shown in figure with initial velocities $v_A = 5\,\text{m/s}, v_B = 2\,\text{m/s}, v_C = 4\,\text{m/s}$. If we wait sufficiently long for elastic collision to happen, then $v_A = 4\,\text{m/s}, v_B = 2\,\text{m/s}, v_C = 5\,\text{m/s}$ will be the final velocities.

Reason (R): In an elastic collision between identical masses, two objects exchange their velocities.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) (A) is false but (R) is true

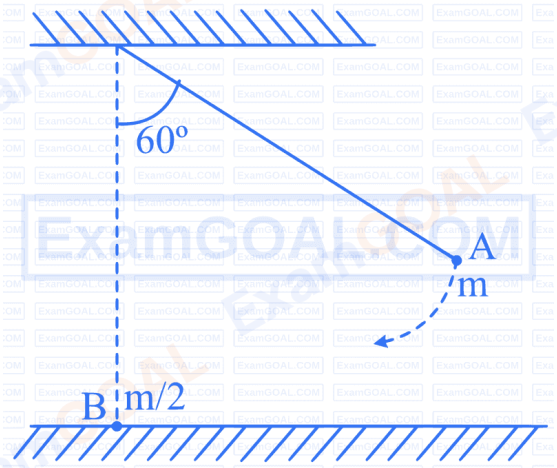
(D) (A) is true but (R) is false

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Q41. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

As shown below, bob A of a pendulum having massless string of length 'R' is released from 60° to the vertical. It hits another bob B of half the mass that is at rest on a frictionless table in the center. Assuming elastic collision, the magnitude of the velocity of bob A after the collision will be (take g as acceleration due to gravity.)

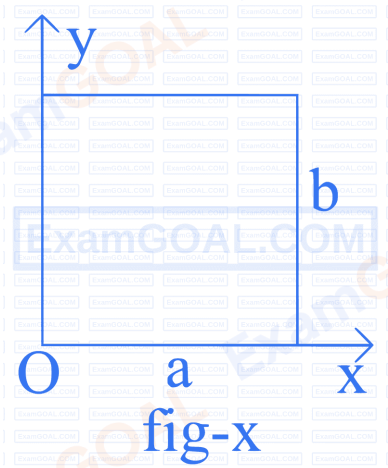


(A) $\frac{4}{3}\sqrt{Rg}$ (B) $\frac{1}{3}\sqrt{Rg}$ (C) $\sqrt{Rg}$ (D) $\frac{2}{3}\sqrt{Rg}$

Q42. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

The center of mass of a thin rectangular plate (fig - x) with sides of length a and b, whose mass per unit area (σ) varies as $\sigma = \frac{\sigma_0 x}{ab}$ (where σ_0 is a constant), would be _____.



(A) $\left(\frac{2}{3}a, \frac{2}{3}b\right)$ (B) $\left(\frac{2}{3}a, \frac{b}{2}\right)$

(C) $\left(\frac{1}{3}a, \frac{b}{2}\right)$ (D) $\left(\frac{a}{2}, \frac{b}{2}\right)$

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Q43. JEE Main 2025 (Online) 23rd January Morning Shift

Consider a circular disc of radius 20 cm with centre located at the origin. A circular hole of radius 5 cm is cut from this disc in such a way that the edge of the hole touches the edge of the disc. The distance of centre of mass of residual or remaining disc from the origin will be

(A) 1.5 cm (B) 2.0 cm (C) 0.5 cm (D) 1.0 cm

Open solution

Q44. JEE Main 2024 (Online) 8th April Morning Shift

A stationary particle breaks into two parts of masses m_A and m_B which move with velocities v_A and v_B respectively. The ratio of their kinetic energies $(K_B : K_A)$ is :

(A) $v_B : v_A$ (B) $1 : 1$ (C) $m_B v_B : m_A v_A$ (D) $m_B : m_A$

Open solution

Q45. JEE Main 2024 (Online) 31st January Morning Shift

An artillery piece of mass M_1 fires a shell of mass M_2 horizontally. Instantaneously after the firing, the ratio of kinetic energy of the artillery and that of the shell is:

(A) $M_1/(M_1 + M_2)$ (B) $\frac{M_2}{M_1}$ (C) $\frac{M_1}{M_2}$ (D) $M_2/(M_1 + M_2)$

Open solution

Q46. JEE Main 2024 (Online) 30th January Morning Shift

A spherical body of mass $100\ \mathrm{g}$ is dropped from a height of $10\ \mathrm{m}$ from the ground. After hitting the ground, the body rebounds to a height of $5\ \mathrm{m}$. The impulse of force imparted by the ground to the body is given by : (given, $\mathrm{g} = 9.8\ \mathrm{m/s^2}$)

(A) $43.2\ \mathrm{kg\ ms^{-1}}$ (B) $2.39\ \mathrm{kg\ ms^{-1}}$

(C) $4.32\ \mathrm{kg\ ms^{-1}}$ (D) $23.9\ \mathrm{kg\ ms^{-1}}$

Open solution

Q47. JEE Main 2024 (Online) 27th January Morning Shift

Two bodies of mass $4\ \mathrm{g}$ and $25\ \mathrm{g}$ are moving with equal kinetic energies. The ratio of magnitude of their linear momentum is :

(A) $3 : 5$ (B) $5 : 4$ (C) $2 : 5$ (D) $4 : 5$

Open solution

Q48. JEE Main 2024 (Online) 27th January Morning Shift

A body of mass $1000\ \mathrm{kg}$ is moving horizontally with a velocity $6\ \mathrm{m/s}$. If $200\ \mathrm{kg}$ extra mass is added, the final velocity (in $\mathrm{m/s}$) is:

(A) 6 (B) 2 (C) 3 (D) 5

Open solution

Q49. JEE Main 2023 (Online) 13th April Morning Shift

A bullet of $10\ \mathrm{g}$ leaves the barrel of gun with a velocity of $600\ \mathrm{m/s}$. If the barrel of gun is $50\ \mathrm{cm}$ long and mass of gun is $3\ \mathrm{kg}$, then value of impulse supplied to the gun will be :

(A) 12 Ns (B) 3 Ns (C) 6 Ns (D) 36 Ns

Open solution

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Q50. JEE Main 2023 (Online) 11th April Morning Shift

An average force of $125\ \mathrm{N}$ is applied on a machine gun firing bullets each of mass $10\ \mathrm{g}$ at the speed of $250\ \mathrm{m/s}$ to keep it in position. The number of bullets fired per second by the machine gun is :

(A) 25 (B) 50 (C) 5 (D) 100

Open solution

Q51. JEE Main 2023 (Online) 10th April Morning Shift

A particle of mass m moving with velocity v collides with a stationary particle of mass 2m. After collision, they stick together and continue to move together with velocity

(A) v (B) $\frac{v}{3}$ (C) $\frac{v}{4}$ (D) $\frac{v}{2}$

Open solution

Q52. JEE Main 2023 (Online) 31st January Morning Shift

100 balls each of mass m moving with speed v simultaneously strike a wall normally and reflected back with same speed, in time $\mathrm{t\ s}$. The total force exerted by the balls on the wall is

(A) $\frac{200mv}{t}$ (B) $\frac{100mv}{t}$ (C) $\frac{mv}{100t}$ (D) $200mvt$

Open solution

Q53. JEE Main 2023 (Online) 30th January Evening Shift

A machine gun of mass $10\ \mathrm{kg}$ fires $20\ \mathrm{g}$ bullets at the rate of 180 bullets per minute with a speed of $100\ \mathrm{m\ s^{-1}}$ each. The recoil velocity of the gun is

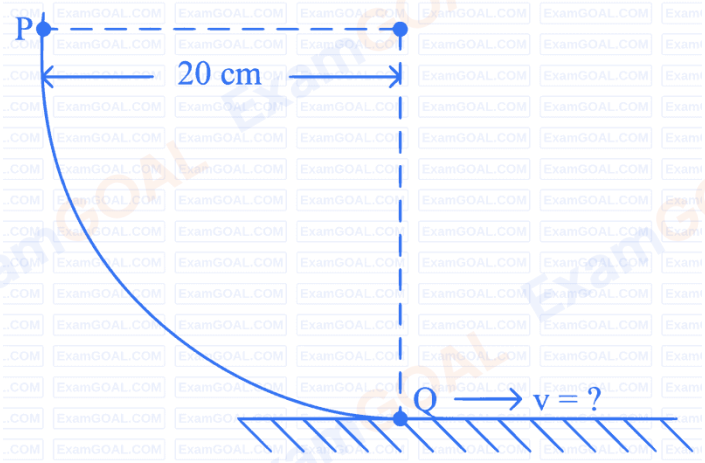
(A) $0.02\ \mathrm{m/s}$ (B) $1.5\ \mathrm{m/s}$ (C) $2.5\ \mathrm{m/s}$ (D) $0.6\ \mathrm{m/s}$

Open solution

Q54. JEE Main 2023 (Online) 30th January Morning Shift

As per the given figure, a small ball P slides down the quadrant of a circle and hits the other ball Q of equal mass which is initially at rest. Neglecting the effect of friction and assume the collision to be elastic, the velocity of ball Q after collision will be :

($g = 10\ m/s^2$)



(A) 0.25 m/s (B) 4 m/s (C) 0 (D) 2 m/s

Open solution

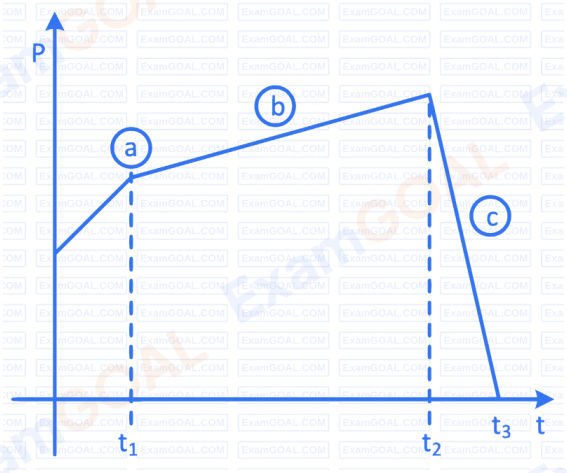
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Q55. JEE Main 2023 (Online) 30th January Morning Shift

The figure represents the momentum time $(\mathrm{p}-\mathrm{t})$ curve for a particle moving along an axis under the influence of the force. Identify the regions on the graph where the magnitude of the force is maximum and minimum respectively?

Open solution

If $(t_3 - t_2) < t_1$



(A) b and c (B) c and a (C) a and b (D) c and b

Q56. JEE Main 2023 (Online) 30th January Morning Shift

A ball of mass $200\ \mathrm{g}$ rests on a vertical post of height $20\ \mathrm{m}$. A bullet of mass $10\ \mathrm{g}$, travelling in horizontal direction, hits the centre of the ball. After collision both travels independently. The ball hits the ground at a distance $30\ \mathrm{m}$ and the bullet at a distance of $120\ \mathrm{m}$ from the foot of the post. The value of initial velocity of the bullet will be (if $g = 10\ \mathrm{m/s^2}$) :

Open solution

(A) 120 m/s (B) 360 m/s (C) 400 m/s (D) 60 m/s

Q57. JEE Main 2022 (Online) 29th July Evening Shift

If momentum of a body is increased by 20%, then its kinetic energy increases by

Open solution

(A) 36% (B) 40% (C) 44% (D) 48%

Q58. JEE Main 2022 (Online) 29th July Morning Shift

Two bodies of mass $1\ \mathrm{kg}$ and $3\ \mathrm{kg}$ have position vectors $\hat{i} + 2\hat{j} + \hat{k}$ and $-3\hat{i} - 2\hat{j} + \hat{k}$ respectively. The magnitude of position vector of centre of mass of this system will be similar to the magnitude of vector :

Open solution

(A) $\hat{i} + 2\hat{j} + \hat{k}$ (B) $-3\hat{i} - 2\hat{j} + \hat{k}$ (C) $-2\hat{i} + 2\hat{k}$ (D) $2\hat{i} - \hat{j} + 2\hat{k}$

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Q59. JEE Main 2022 (Online) 28th July Morning Shift

In two different experiments, an object of mass $5\ \text{kg}$ moving with a speed of $25\ \text{ms}^{-1}$ hits two different walls and comes to rest within (i) 3 second, (ii) 5 seconds, respectively. Choose the correct option out of the following :

(A) Impulse and average force acting on the object will be same for both the cases.

(B) Impulse will be same for both the cases but the average force will be different.

(C) Average force will be same for both the cases but the impulse will be different.

(D) Average force and impulse will be different for both the cases.

Open solution

Q60. JEE Main 2022 (Online) 26th July Evening Shift

A ball of mass $0.15\ \text{kg}$ hits the wall with its initial speed of $12\ \text{ms}^{-1}$ and bounces back without changing its initial speed. If the force applied by the wall on the ball during the contact is $100\ \text{N}$, calculate the time duration of the contact of ball with the wall.

(A) 0.018 s (B) 0.036 s (C) 0.009 s (D) 0.072 s

Open solution

Q61. JEE Main 2022 (Online) 26th July Evening Shift

A body of mass $8\ \text{kg}$ and another of mass $2\ \text{kg}$ are moving with equal kinetic energy. The ratio of their respective momentum will be :

(A) 1 : 1 (B) 2 : 1 (C) 1 : 4 (D) 4 : 1

Open solution

Q62. JEE Main 2022 (Online) 25th July Evening Shift

Two billiard balls of mass 0.05 kg each moving in opposite directions with 10 ms^{-1} collide and rebound with the same speed. If the time duration of contact is t = 0.005 s, then what is the force exerted on the ball due to each other?

(A) 100 N (B) 200 N (C) 300 N (D) 400 N

Open solution

Q63. JEE Main 2022 (Online) 30th June Morning Shift

Two bodies A and B of masses 5 kg and 8 kg are moving such that the momentum of body B is twice that of the body A. The ratio of their kinetic energies will be :

(A) 4 : 5 (B) 2 : 5 (C) 5 : 4 (D) 5 : 2

Open solution

Q64. JEE Main 2022 (Online) 29th June Morning Shift

A body of mass M at rest explodes into three pieces, in the ratio of masses 1 : 1 : 2. Two smaller pieces fly off perpendicular to each other with velocities of 30 ms^{-1} and 40 ms^{-1} respectively. The velocity of the third piece will be :

(A) 15 ms^{-1} (B) 25 ms^{-1} (C) 35 ms^{-1} (D) 50 ms^{-1}

Open solution

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Q65. JEE Main 2022 (Online) 27th June Morning Shift

Two blocks of masses 10 kg and 30 kg are placed on the same straight line with coordinates (0, 0) cm and (x, 0) cm respectively. The block of 10 kg is moved on the same line through a distance of 6 cm towards the other block. The distance through which the block of 30 kg must be moved to keep the position of centre of mass of the system unchanged is :

(A) 4 cm towards the 10 kg block
(B) 2 cm away from the 10 kg block
(C) 2 cm towards the 10 kg block
(D) 4 cm away from the 10 kg block

Open solution

Q66. JEE Main 2022 (Online) 27th June Morning Shift

What percentage of kinetic energy of a moving particle is transferred to a stationary particle when it strikes the stationary particle of 5 times its mass?

(Assume the collision to be head-on elastic collision)

(A) 50.0%
(B) 66.6%
(C) 55.6%
(D) 33.3%

Open solution

Q67. JEE Main 2022 (Online) 26th June Morning Shift

An object is thrown vertically upwards. At its maximum height, which of the following quantity becomes zero?

(A) Momentum
(B) Potential Energy
(C) Acceleration
(D) Force

Open solution

Q68. JEE Main 2021 (Online) 31st August Morning Shift

A body of mass M moving at speed V_0 collides elastically with a mass 'm' at rest. After the collision, the two masses move at angles θ_1 and θ_2 with respect to the initial direction of motion of the body of mass M. The largest possible value of the ratio M/m, for which the angles θ_1 and θ_2 will be equal, is :

(A) 4
(B) 1
(C) 3
(D) 2

Open solution

Q69. JEE Main 2021 (Online) 26th August Morning Shift

The initial mass of a rocket is 1000 kg. Calculate at what rate the fuel should be burnt so that the rocket is given an acceleration of 20 ms^{-2}. The gases come out at a relative speed of 500 ms^{-1} with respect to the rocket : [Use g = 10 m/s^2]

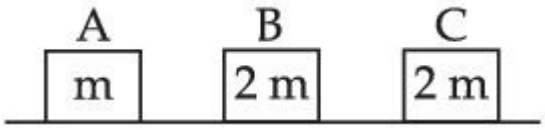
(A) 6.0×10^2 kg s^{-1}
(B) 500 kg s^{-1}
(C) 10 kg s^{-1}
(D) 60 kg s^{-1}

Open solution

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Q70. JEE Main 2021 (Online) 27th July Morning Shift

Three objects A, B and C are kept in a straight line on a frictionless horizontal surface. The masses of A, B and C are m, 2m and 2m respectively. A moves towards B with a speed of 9 m/s and makes an elastic collision with it. Thereafter B makes a completely inelastic collision with C. All motions occur along same straight line. The final speed of C is :

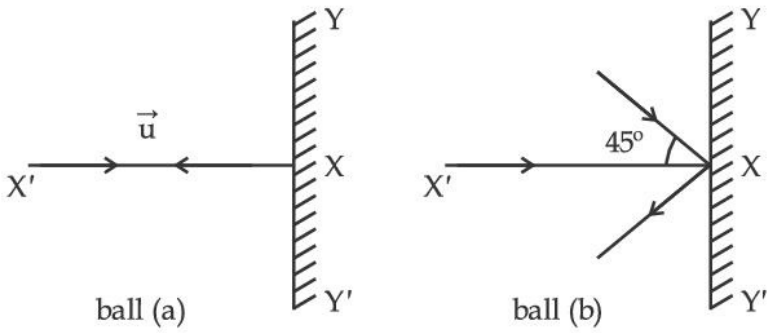


Open solution

(A) 6 m/s (B) 9 m/s (C) 4 m/s (D) 3 m/s

Q71. JEE Main 2021 (Online) 25th July Morning Shift

Two billiard balls of equal mass 30g strike a rigid wall with same speed of 108 kmph (as shown) but at different angles. If the balls get reflected with the same speed then the ratio of the magnitude of impulses imparted to ball 'a' and ball 'b' by the wall along 'X' direction is :



Open solution

(A) 1 : 1 (B) $\sqrt{2}$: 1 (C) 2 : 1 (D) 1 : $\sqrt{2}$

Q72. JEE Main 2021 (Online) 22th July Evening Shift

A bullet of '4 g' mass is fired from a gun of mass 4 kg. If the bullet moves with the muzzle speed of 50 ms^{-1}, the impulse imparted to the gun and velocity of recoil of gun are :

Open solution

(A) 0.2 kg ms^{-1}, 0.1 ms^{-1} (B) 0.4 kg ms^{-1}, 0.05 ms^{-1}

(C) 0.2 kg ms^{-1}, 0.05 ms^{-1} (D) 0.4 kg ms^{-1}, 0.1 ms^{-1}

Q73. JEE Main 2021 (Online) 20th July Evening Shift

If the Kinetic energy of a moving body becomes four times its initial Kinetic energy, then the percentage change in its momentum will be :

Open solution

(A) 100% (B) 200% (C) 300% (D) 400%

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Q74. JEE Main 2021 (Online) 18th March Evening Shift

An object of mass m_1 collides with another object of mass m_2, which is at rest. After the collision the objects move with equal speeds in opposite direction. The ratio of the masses $m_2 : m_1$ is :

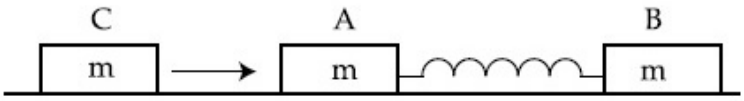
Open solution

(A) 1 : 1 (B) 3 : 1 (C) 2 : 1 (D) 1 : 2

Q75. JEE Main 2021 (Online) 17th March Evening Shift

Two identical blocks A and B each of mass m resting on the smooth horizontal floor are connected by a light spring of natural length L and spring constant K. A third block C of mass m moving with a speed v along the line joining A and B collides with A. The maximum compression in the spring is

Open solution



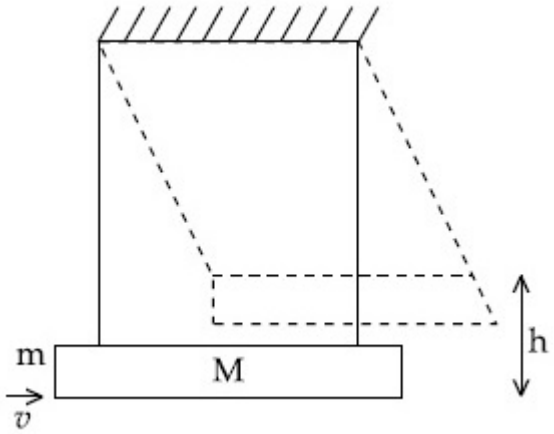
(A) $\sqrt{\frac{mv}{K}}$ (B) $\sqrt{\frac{m}{2K}}$ (C) $\sqrt{\frac{mv}{2K}}$ (D) $v\sqrt{\frac{m}{2K}}$

Q76. JEE Main 2021 (Online) 16th March Evening Shift

A large block of wood of mass M = 5.99 kg is hanging from two long massless cords. A bullet of mass m = 10 g is fired into the block and gets embedded in it. The (block + bullet) then swing upwards, their centre of mass rising a vertical distance h = 9.8 cm before the (block + bullet) pendulum comes momentarily to rest at the end of its arc. The speed of the bullet just before collision is :

(take g = 9.8 ms^{-2})

Open solution



(A) 831.4 m/s (B) 811.4 m/s (C) 841.4 m/s (D) 821.4 m/s

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Q77. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Body 'P' having mass M moving with speed 'u' has head-on collision elastically with another body 'Q' having mass 'm' initially at rest. If m << M, body 'Q' will have a maximum speed equal to '2u' after collision.

Reason R : During elastic collision, the momentum and kinetic energy are both conserved.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is correct but R is not correct.

(B) A is not correct but R is correct.

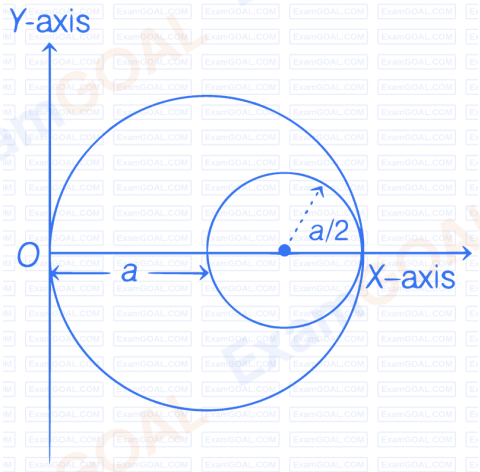
(C) Both A and R are correct and R is the correct explanation of A.

(D) Both A and R are correct but R is NOT the correct explanation of A.

Q78. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

A circular hole of radius $\left(\frac{a}{2}\right)$ is cut out of a circular disc of radius 'a' as shown in figure. The centroid of the remaining circular portion with respect to point 'O' will be :



(A) $\frac{1}{6}a$ **(B)** $\frac{2}{3}a$ **(C)** $\frac{5}{6}a$ **(D)** $\frac{10}{11}a$

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Q79. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

Particle A of mass m_1 moving with velocity $\left(\sqrt{3}\hat{i}+\hat{j}\right)ms^{-1}$ collides with another particle B of mass m_2 which is at rest initially. Let $\overrightarrow{V_1}$ and $\overrightarrow{V_2}$ be the velocities of particles A and B after collision respectively. If $m_1 = 2m_2$ and after collision $\overrightarrow{V_1}=\left(\hat{i}+\sqrt{3}\hat{j}\right)$, the angle between $\overrightarrow{V_1}$ and $\overrightarrow{V_2}$ is :

(A) 105° (B) 15° (C) -45° (D) 60°

Q80. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

A spaceship in space sweeps stationary interplanetary dust. As a result, its mass increases at a rate $\frac{dM(t)}{dt} = bv^2(t)$, where v(t) is its instantaneous velocity. The instantaneous acceleration of the satellite is :

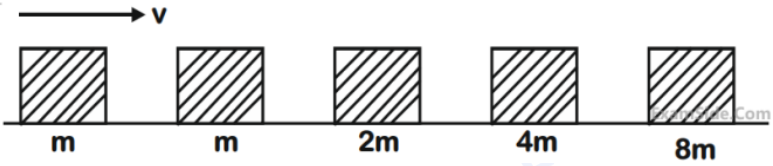
(A) $-bv^3(t)$ (B) $-\frac{2bv^3}{M(t)}$

(C) $-\frac{bv^3}{M(t)}$ (D) $-\frac{bv^3}{2M(t)}$

Q81. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

Blocks of masses m, 2m, 4m and 8m are arranged in a line on a frictionless floor. Another block of mass m, moving with speed v along the same line (see figure) collides with mass m in perfectly inelastic manner. All the subsequent collisions are also perfectly inelastic. By the time the last block of mass 8m starts moving the total energy loss is p% of the original energy. Value of 'p' is close to :



(A) 37 (B) 77 (C) 87 (D) 94

Q82. JEE Main 2020 (Online) 3rd September Evening Slot

Open solution

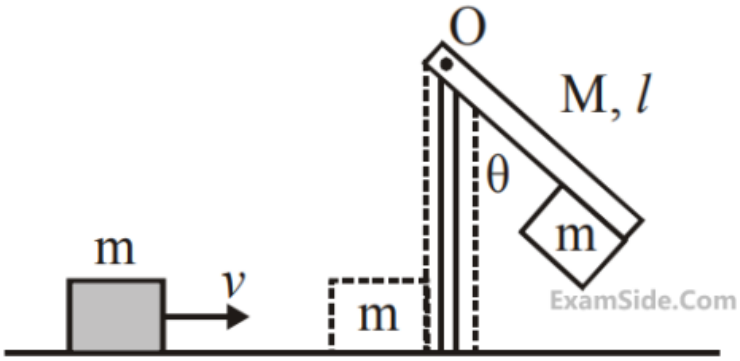
A block of mass 1.9 kg is at rest at the edge of a table, of height 1 m. A bullet of mass 0.1 kg collides with the block and sticks to it. If the velocity of the bullet is 20 m/s in the horizontal direction just before the collision then the kinetic energy just before the combined system strikes the floor, is [Take g = 10 m/s^2 . Assume there is no rotational motion and loss of energy after the collision is negligible.]

(A) 23 J (B) 21 J (C) 20 J (D) 19 J

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Q83. JEE Main 2020 (Online) 3rd September Morning Slot

A block of mass m = 1 kg slides with velocity v = 6 m/s on a frictionless horizontal surface and collides with a uniform vertical rod and sticks to it as shown. The rod is pivoted about O and swings as a result of the collision making angle θ before momentarily coming to rest. If the rod has mass M = 2 kg, and length l = 1 m, the value of θ is approximately :
(take g = 10 m/s^2)



Open solution

(A) 63° (B) 69° (C) 55° (D) 49°

Q84. JEE Main 2020 (Online) 2nd September Morning Slot

A particle of mass m with an initial velocity $u\hat{i}$ collides perfectly elastically with a mass 3 m at rest. It moves with a velocity $v\hat{j}$ after collision, then, v is given by :

(A) $v = \sqrt{\frac{2}{3}}u$ (B) $v = \frac{u}{\sqrt{3}}$ (C) $v = \frac{u}{\sqrt{2}}$ (D) $v = \frac{1}{\sqrt{6}}u$

Open solution

Q85. JEE Main 2020 (Online) 9th January Evening Slot

A rod of length L has non-uniform linear mass density given by $\rho(x) = a + b\left(\frac{x}{L}\right)^2$, where a and b are constants and $0 \le x \le L$. The value of x for the centre of mass of the rod is at :

(A) $\frac{3}{2}\left(\frac{a+b}{2a+b}\right)L$ (B) $\frac{4}{3}\left(\frac{a+b}{2a+3b}\right)L$
(C) $\frac{3}{4}\left(\frac{2a+b}{3a+b}\right)L$ (D) $\frac{3}{2}\left(\frac{2a+b}{3a+b}\right)L$

Open solution

Q86. JEE Main 2020 (Online) 9th January Evening Slot

A particle of mass m is projected with a speed u from the ground at an angle $\theta = \frac{\pi}{3}$ w.r.t. horizontal (x-axis). When it has reached its maximum height, it collides completely inelastically with another particle of the same mass and velocity $u\hat{i}$. The horizontal distance covered by the combined mass before reaching the ground is:

(A) $2\sqrt{2}\frac{u^2}{g}$ (B) $\frac{3\sqrt{3}}{8}\frac{u^2}{g}$
(C) $\frac{3\sqrt{2}}{4}\frac{u^2}{g}$ (D) $\frac{5}{8}\frac{u^2}{g}$

Open solution

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Q87. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

Two particles of equal mass m have respective initial velocities $u\hat{i}$ and $u\left(\frac{\hat{i}+\hat{j}}{2}\right)$.

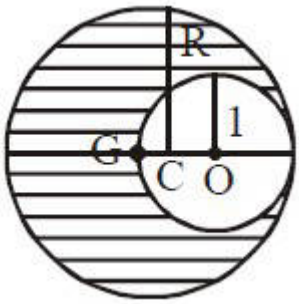
They collide completely inelastically. The energy lost in the process is :

(A) $\frac{1}{3}mu^2$ (B) $\frac{1}{8}mu^2$ (C) $\frac{3}{4}mu^2$ (D) $\sqrt{\frac{2}{3}}mu^2$

Q88. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

As shown in figure, when a spherical cavity (centered at O) of radius 1 is cut out of a uniform sphere of radius R (centered at C), the centre of mass of remaining (shaded) part of sphere is at G, i.e, on the surface of the cavity. R can be determined by the equation :



(A) $(R^2 + R - 1)(2 - R) = 1$

(B) $(R^2 - R - 1)(2 - R) = 1$

(C) $(R^2 - R + 1)(2 - R) = 1$

(D) $(R^2 + R + 1)(2 - R) = 1$

Q89. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

A particle of mass m is dropped from a height h above the ground. At the same time another particle of the same mass is thrown vertically upwards from the ground with a speed of $\sqrt{2gh}$. If they collide head-on completely inelastically, the time taken for the combined mass to reach the ground, in units of $\sqrt{\frac{h}{g}}$ is :

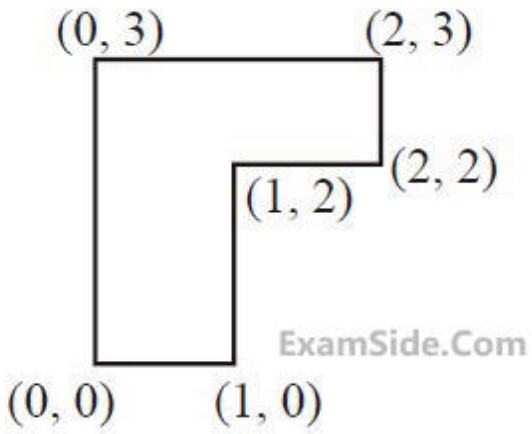
(A) $\sqrt{\frac{1}{2}}$ (B) $\frac{1}{2}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\sqrt{\frac{3}{4}}$

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Q90. JEE Main 2020 (Online) 8th January Morning Slot

Open solution

The coordinates of centre of mass of a uniform flag shaped lamina (thin flat plate) of mass 4kg. (The coordinates of the same are shown in figure) are :

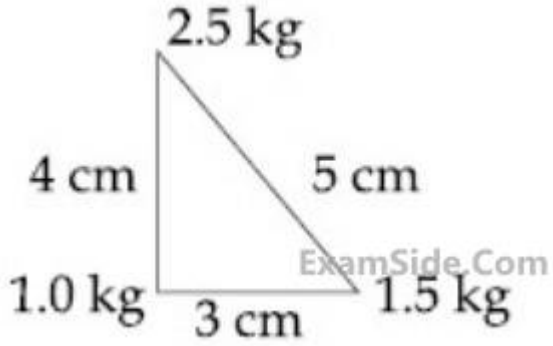


(A) (0.75m, 1.75m) (B) (1m, 1.75m) (C) (0.75m, 0.75m) (D) (1.25m, 1.50m)

Q91. JEE Main 2020 (Online) 7th January Morning Slot

Open solution

Three point particles of masses 1.0 kg, 1.5 kg and 2.5 kg are placed at three corners of a right angle triangle of sides 4.0 cm, 3.0 cm and 5.0 cm as shown in the figure. The center of mass of the system is at a point:



(A) 2.0 cm right and 0.9 cm above 1 kg mass (B) 0.9 cm right and 2.0 cm above 1 kg mass

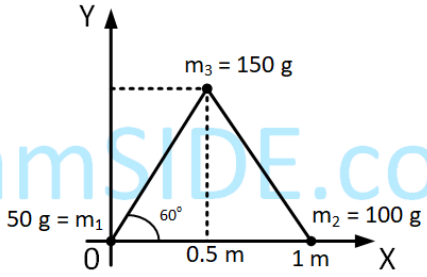
(C) 0.6 cm right and 2.0 cm above 1 kg mass (D) 1.5 cm right and 1.2 cm above 1 kg mass

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Q92. JEE Main 2019 (Online) 12th April Evening Slot

Three particles of masses 50 g, 100 g and 150 g are placed at the vertices of an equilateral triangle of side 1 m (as shown in the figure). The (x, y) coordinates of the centre of mass will be :

Open solution



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(A) $\left(\frac{\sqrt{3}}{4}m, \frac{5}{12}m\right)$

(B) $\left(\frac{7}{12}m, \frac{\sqrt{3}}{8}m\right)$

(C) $\left(\frac{7}{12}m, \frac{\sqrt{3}}{4}m\right)$

(D) $\left(\frac{\sqrt{3}}{8}m, \frac{7}{12}m\right)$

Q93. JEE Main 2019 (Online) 12th April Morning Slot

A man (mass = 50 kg) and his son (mass = 20 kg) are standing on a frictionless surface facing each other. The man pushes his son so that he starts moving at a speed of 0.70 ms^{-1} with respect to the man. The speed of the man with respect to the surface is :

Open solution

(A) 0.28 ms^{-1}

(B) 0.47 ms^{-1}

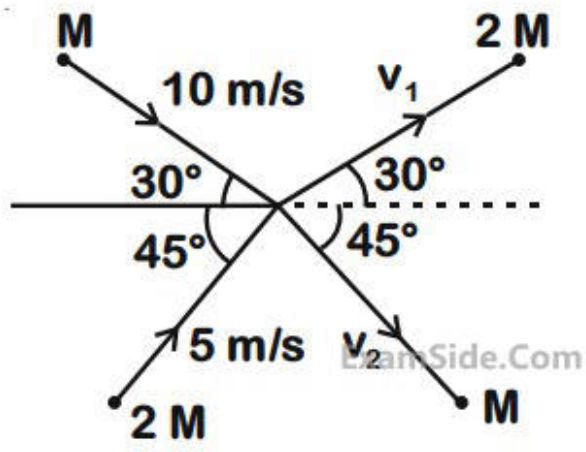
(C) 0.20 ms^{-1}

(D) 0.14 ms^{-1}

Q94. JEE Main 2019 (Online) 10th April Morning Slot

Two particles, of masses M and 2M, moving, as shown, with speeds of 10 m/s and 5 m/s, collide elastically at the origin. After the collision, they move along the indicated directions with speeds u_1 and u_2, respectively. The values of u_1 and u_2 are nearly :

Open solution



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(A) 6.5 m/s and 3.2 m/s

(B) 6.5 m/s and 6.3 m/s

(C) 3.2 m/s and 6.3 m/s

(D) 3.2 m/s and 12.6 m/s

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Q95. JEE Main 2019 (Online) 9th April Evening Slot

A particle of mass 'm' is moving with speed '2v' and collides with a mass '2m' moving with speed 'v' in the same direction. After collision, the first mass is stopped completely while the second one splits into two particles each of mass 'm', which move at angle 45° with respect to the origianl direction. The speed of each of the moving particle will be :-

Open solution

(A) $2\sqrt{2}v$ (B) $v / (2\sqrt{2})$ (C) $v / \sqrt{2}$ (D) $\sqrt{2}v$

Q96. JEE Main 2019 (Online) 9th April Evening Slot

A wedge of mass M = 4m lies on a frictionless plane. A particle of mass m approaches the wedge with speed v. There is no friction between the particle and the plane or between the particle and the wedge. The maximum height climbed by the particle on the wedge is given by :-

Open solution

(A) $\frac{v^2}{g}$ (B) $\frac{2v^2}{7g}$ (C) $\frac{v^2}{2g}$ (D) $\frac{2v^2}{5g}$

Q97. JEE Main 2019 (Online) 9th April Morning Slot

A body of mass 2 kg makes an elastic collision with a second body at rest and continues to move in the original direction but with one fourth of its original speed. What is the mass of the second body ?

Open solution

(A) 1.2 kg (B) 1.0 kg (C) 1.8 kg (D) 1.5 kg

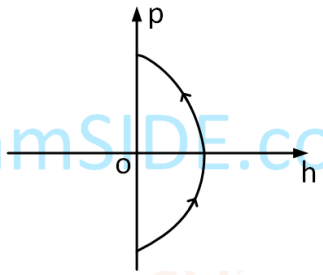
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Q98. JEE Main 2019 (Online) 9th April Morning Slot

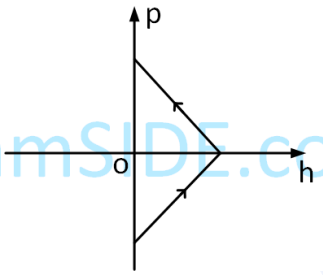
Open solution

A ball is thrown vertically up (taken as +z-axis) from the ground. The correct momentum-height (p-h) diagram is :

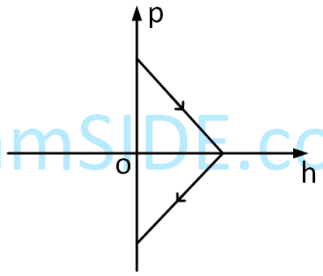
(A)



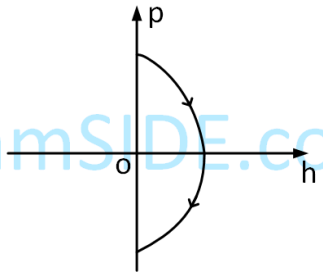
(B)



(C)



(D)

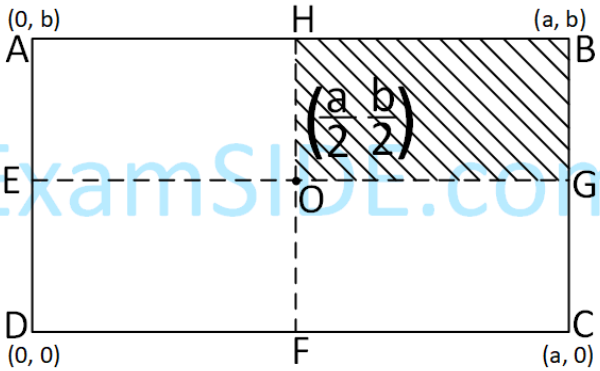


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Q99. JEE Main 2019 (Online) 8th April Evening Slot

Open solution

A uniform rectangular thin sheet ABCD of mass M has length a and breadth b, as shown in the figure. If the shaded portion HBGO is cut-off, the coordinates of the centre of mass of the remaining portion will be :-



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(A) $\left(\frac{5a}{3}, \frac{5b}{3}\right)$ (B) $\left(\frac{2a}{3}, \frac{2b}{3}\right)$
(C) $\left(\frac{5a}{12}, \frac{5b}{12}\right)$ (D) $\left(\frac{3a}{4}, \frac{3b}{4}\right)$

Q100. JEE Main 2019 (Online) 8th April Evening Slot

Open solution

A body of mass m_1 moving with an unknown velocity of $v_1\hat{i}$, undergoes a collinear collision with a body of mass m_2 moving with a velocity $v_2\hat{i}$. After collision, m_1 and m_2 move with velocities of $v_3\hat{i}$ and $v_4\hat{i}$, respectively. If $m_2 = 0.5\ m_1$ and $v_3 = 0.5\ v_1$, then v_1 is :-

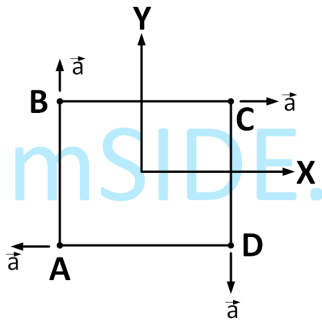
(A) $v_4 - \frac{v_2}{2}$ (B) $v_4 - \frac{v_2}{4}$ (C) $v_4 - v_2$ (D) $v_4 + v_2$

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Q101. JEE Main 2019 (Online) 8th April Morning Slot

Four particles A, B, C and D with masses m_A = m, m_B = 2m, m_C = 3m and m_D = 4m are at the corners of a square. They have accelerations of equal magnitude with directions as shown. The acceleration of the centre of mass of the particles is :

Open solution



ExamSIDE.com

(A) $\frac{a}{5}\left(\hat{i}+\hat{j}\right)$

(B) $\frac{a}{5}\left(\hat{i}-\hat{j}\right)$

(C) $a\left(\hat{i}+\hat{j}\right)$

(D) Zero

Q102. JEE Main 2019 (Online) 12th January Evening Slot

An alpha-particle of mass m suffers 1-dimensional elastic collision with a nucleus at rest of unknown mass. It is scattered directly backwards losing, 64% of its initial kinetic energy. The mass of the nucleus is

Open solution

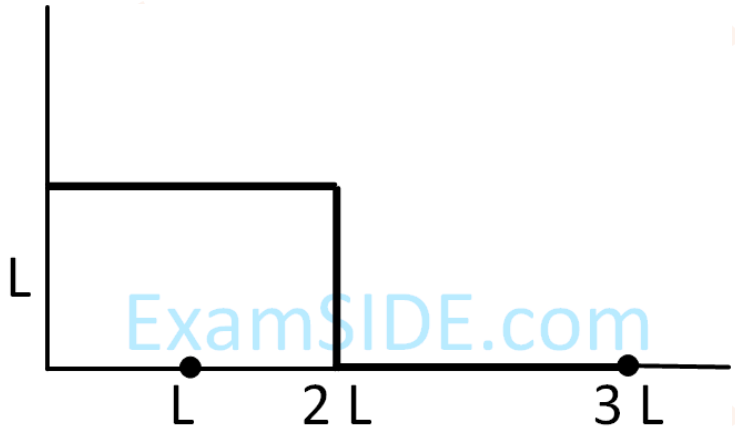
(A) 2m (B) 4m (C) 1.5m (D) 3.5m

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Q103. JEE Main 2019 (Online) 12th January Morning Slot

The position vector of the centre of mass $\vec{r}_{cm}$ of an asymmetric uniform bar of negligible area of cross-section as shown in figure is :

Open solution



(A) $\vec{r}_{cm} = \frac{11}{8}L\,\hat{x} + \frac{8}{8}L\hat{y}$

(B) $\vec{r}_{cm} = \frac{5}{8}L\,\hat{x} + \frac{13}{8}L\hat{y}$

(C) $\vec{r}_{cm} = \frac{13}{8}L\,\hat{x} + \frac{5}{8}L\hat{y}$

(D) $\vec{r}_{cm} = \frac{3}{8}L\,\hat{x} + \frac{11}{8}L\hat{y}$

Q104. JEE Main 2019 (Online) 12th January Morning Slot

A simple pendulum, made of a string of length ℓ and a bob of mass m, is released from a small angle θ_0. It strikes a block of mass M, kept on a horizontal surface at its lowest point of oscillations, elastically. It bounces back and goes up to an angle θ_1. Then M is given by :

Open solution

(A) $\frac{m}{2}\left(\frac{\theta_0+\theta_1}{\theta_0-\theta_1}\right)$

(B) $\frac{m}{2}\left(\frac{\theta_0-\theta_1}{\theta_0+\theta_1}\right)$

(C) $m\left(\frac{\theta_0+\theta_1}{\theta_0-\theta_1}\right)$

(D) $m\left(\frac{\theta_0-\theta_1}{\theta_0+\theta_1}\right)$

Q105. JEE Main 2019 (Online) 11th January Evening Slot

A particle of mass m is moving in a straight line with momentum p. Starting at time t = 0, a force F = kt acts in the same direction on the moving particle during time interval T so that its momentum changes from p to 3p. Here k is a constant. The value of T is :

Open solution

(A) $2\sqrt{\frac{k}{p}}$ (B) $2\sqrt{\frac{p}{k}}$ (C) $\sqrt{\frac{2p}{k}}$ (D) $\sqrt{\frac{2k}{p}}$

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Q106. JEE Main 2019 (Online) 11th January Morning Slot

A body of mass 1 kg falls freely from a height of 100 m, on a platform mass 3 kg which is mounted on a spring having spring constant k = 1.25×10^6 N/m. The body sticks to the platform and the spring's maximum compression is found to be x. Given that g = 10 ms^{-2}, the value of x will be close to :

Open solution

(A) 8 cm (B) 4 cm (C) 40 cm (D) 80 cm

Q107. JEE Main 2019 (Online) 10th January Morning Slot

A piece of wood of mass 0.03 kg is dropped from the top of a 100 m height building. At the same time, a bullet of mass 0.02 kg is fired vertically upward, with a velocity 100 ms^{-1}, from the ground. The bullet gets embedded in the wood. Then the maximum height to which the combined system reaches above the top of the building before falling below is - (g = 10 ms^{-2})

Open solution

(A) 30 m (B) 40 m (C) 20 m (D) 10 m

Q108. JEE Main 2019 (Online) 9th January Morning Slot

Three blocks A, B and C are lying on a smooth horizontal surface, as shown in the figure. A and B have equal masses, m while C has mass M. Block A is given an initial speed v towards B due to which it collides with B perfectly inelastically. The combined mass collides with C, also perfectly inelastically. $\frac{5}{6}$th of the initial kinetic energy is lost in whole process. What is value of M/m ?

Open solution



(A) 5 (B) 2 (C) 4 (D) 3

Q109. JEE Main 2018 (Offline)

In a collinear collision, a particle with an initial speed v_0 strikes a stationary particle of the same mass. If the final total kinetic energy is 50% greater than the original kinetic energy, the magnitude of the relative velocity between the two particles, after collision, is :

Open solution

(A) $\frac{v_0}{\sqrt{2}}$ (B) $\frac{v_0}{4}$ (C) $\sqrt{2}v_0$ (D) $\frac{v_0}{2}$

Q110. JEE Main 2018 (Offline)

It is found that if a neutron suffers an elastic collinear collision with deuterium at rest, fractional loss of its energy is p_d; while for its similar collision with carbon nucleus at rest, fractional loss of energy is p_c. The values of p_d and p_c are respectively :

Open solution

(A) (0, 1) (B) (0.89, 0.28) (C) (0.28, 0.89) (D) (0, 0)

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Q111. JEE Main 2018 (Online) 15th April Evening Slot

A proton of mass m collides elastically with a particle of unknown mass at rest. After the collision, the proton and the unknown particle are seen moving at an angle of 90° with respect to each other. The mass of unknown particle is :

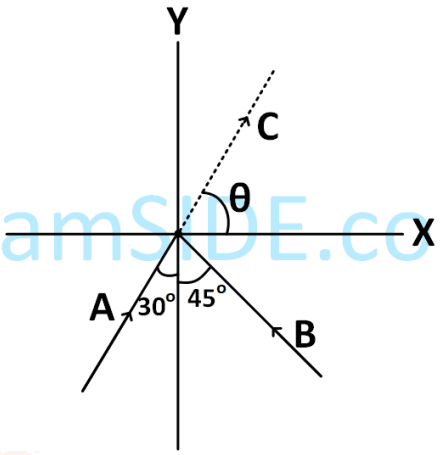
Open solution

(A) $\frac{m}{2}$ (B) m (C) $\frac{m}{\sqrt{3}}$ (D) 2 m

Q112. JEE Main 2017 (Online) 9th April Morning Slot

Two particles A and B of equal mass M are moving with the same speed v as shown in the figure. They collide completely inelastically and move as a single particle C. The angle θ that the path of C makes with the X-axis is given by :

Open solution



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(A) $\tan\theta = \frac{\sqrt{3}+\sqrt{2}}{1-\sqrt{2}}$

(B) $\tan\theta = \frac{\sqrt{3}-\sqrt{2}}{1-\sqrt{2}}$

(C) $\tan\theta = \frac{1-\sqrt{2}}{\sqrt{2}\left(1+\sqrt{3}\right)}$

(D) $\tan\theta = \frac{1-\sqrt{3}}{1+\sqrt{2}}$

Q113. JEE Main 2016 (Online) 10th April Morning Slot

A neutron moving with a speed 'v' makes a head on collision with a stationary hydrogen atom in ground state. The minimum kinetic energy of the neutron for which inelastic collision will take place is :

Open solution

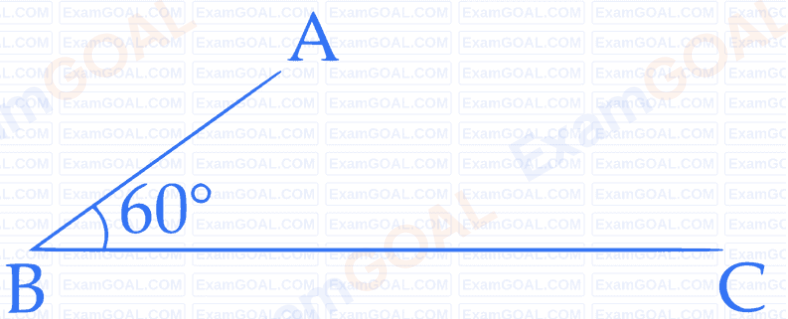
(A) 10.2 eV (B) 16.8 eV (C) 12.1 eV (D) 20.4 eV

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Q114. JEE Main 2016 (Online) 10th April Morning Slot

In the figure shown ABC is a uniform wire. If centre of mass of wire lies vertically below point A, then $\frac{BC}{AB}$ is close to :

Open solution

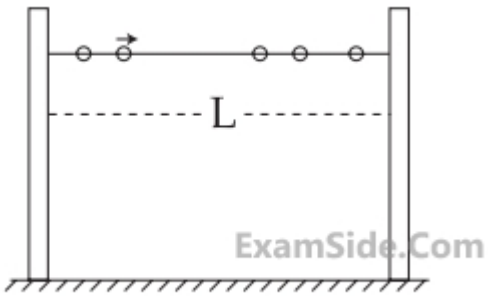


(A) 1.85 (B) 1.37 (C) 1.5 (D) 3

Q115. JEE Main 2015 (Online) 11th April Morning Slot

A large number (n) of identical beads, each of mass m and radius r are strung on a thin smooth rigid horizontal rod of length L (L >> r) and are at rest at random positions. The rod is mounted between two rigid supports (see figure). If one of the beads is now given a speed v, the average force experienced by each support after a long time is (assume all collisions are elastic) :

Open solution



(A) $\frac{mv^2}{L-nr}$ (B) $\frac{mv^2}{L-2nr}$
(C) $\frac{mv^2}{2(L-nr)}$ (D) zero

Q116. JEE Main 2015 (Online) 11th April Morning Slot

A uniform thin rod AB of length L has linear mass density μ (x) = a + $\frac{bx}{L}$, where x is measured from A. If the CM of the rod lies at a distance of $\left(\frac{7}{12}L\right)$ from A, then a and b are related as :

Open solution

(A) a = b (B) a = 2b (C) 2a = b (D) 3a = 2b

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Q117. JEE Main 2015 (Online) 10th April Morning Slot

A block of mass m = 10 kg rests on a horizontal table. The coefficient of friction between the block and the table is 0.05. When hit by a bullet of mass 50 g moving with speed v, that gets embedded in it, the block moves and comes to stop after moving a distance of 2 m on the table.

If a freely falling object were to acquire speed $\frac{v}{10}$ after being dropped from height H, then neglecting energy losses and taking g = 10 ms^{-2}, the value of H is close to :

(A) 0.2 km (B) 0.3 km (C) 0.4 km (D) 0.5 km

Open solution

Q118. JEE Main 2015 (Offline)

A particle of mass m moving in the x direction with speed $2v$ is hit by another particle of mass $2m$ moving in the y direction with speed v. If the collision is perfectly inelastic, the percentage loss in the energy during the collision is close to:

(A) 56% (B) 62% (C) 44% (D) 50%

Open solution

Q119. JEE Main 2015 (Offline)

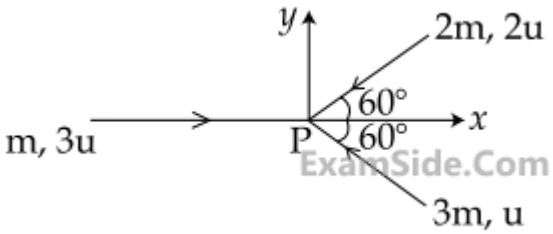
Distance of the center of mass of a solid uniform cone from its vertex is z_0. If the radius of its base is R and its height is h then z_0 is equal to :

(A) $\frac{5h}{8}$ (B) $\frac{3h^2}{8R}$ (C) $\frac{h^2}{4R}$ (D) $\frac{3h}{4}$

Open solution

Q120. JEE Main 2014 (Online) 12th April Morning Slot

Three masses m, 2m and 3m are moving in x-y plane with speed 3u, 2u, and u respectively as shown in Figure. The three masses collide at the same point at P and stick together. The velocity of resulting mass will be :



(A) $\frac{u}{12}\left(\hat{i}+\sqrt{3}\hat{j}\right)$ (B) $\frac{u}{12}\left(\hat{i}-\sqrt{3}\hat{j}\right)$

(C) $\frac{u}{12}\left(-\hat{i}+\sqrt{3}\hat{j}\right)$ (D) $\frac{u}{12}\left(-\hat{i}-\sqrt{3}\hat{j}\right)$

Open solution

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Q121. JEE Main 2014 (Online) 12th April Morning Slot

A 4 g bullet is fired horizontally with a speed of 300 m/s into 0.8 kg block of wood at rest on a table. If the coefficient of friction between the block and the table is 0.3, how far will block slide approximately ?

Open solution

(A) 0.19 m (B) 0.379 m (C) 0.569 m (D) 0.758 m

Q122. JEE Main 2014 (Online) 11th April Morning Slot

A thin bar of length L has a mass per unit length λ, that increases linearly with distance from one end. If its total mass is M and its mass per unit length at the lighter end is λ_o, then the distance of the center of mass from the lighter end is :

Open solution

(A) $\frac{L}{2} - \frac{\lambda_o L^2}{4M}$ (B) $\frac{L}{3} + \frac{\lambda_o L^2}{8M}$

(C) $\frac{L}{3} + \frac{\lambda_o L^2}{4M}$ (D) $\frac{2L}{3} - \frac{\lambda_o L^2}{6M}$

Q123. JEE Main 2013 (Online) 23th April Morning Slot

A boy of mass 20 kg is standing on a 80 kg free to move long cart. There is negligible friction between cart and ground. Initially, the boy is standing 25 m from a wall. If he walks 10 m on the cart towards the wall, then the final distance of the boy from the wall will be

Open solution

(A) 15 m (B) 12.5 m (C) 15.5 m (D) 17 m

Q124. JEE Main 2013 (Online) 23th April Morning Slot

A projectile of mass M is fired so that the horizontal range is 4 km. At the highest point the projectile explodes in two parts of masses M/4 and 3M/4 respectively and the heavier part starts falling down vertically with zero initial speed. The horizontal range (distance from point of firing) of the lighter part is :

Open solution

(A) 16 km (B) 1 km (C) 10 km (D) 2 km

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Q125. JEE Main 2013 (Offline)

This question has statement I and statement II. Of the four choices given after the statements, choose the one that best describes the two statements.

Statement - I: A point particle of mass m moving with speed v collides with stationary point particle of mass M. If the maximum energy loss possible is given as $f\left(\frac{1}{2}mv^2\right)$, then $f = \left(\frac{m}{M+m}\right)$.

Statement - II: Maximum energy loss occurs when the particles get stuck together as a result of the collision.

(A) Statement - I is true, Statement - II is true; Statement - II is the correct explanation of Statement - I.

(B) Statement - I is true, Statement - II is true; Statement - II is not the correct explanation of Statement - I.

(C) Statement - I is true, Statement - II is false

(D) Statement - I is false, Statement - II true.

Open solution

Q126. AIEEE 2012 (Online) 26th May

A satellite moving with velocity v in a force free space collects stationary interplanetary dust at a rate of $\frac{dM}{dt} = \alpha v$ where M is the mass (of satellite + dust) at that instant. The instantaneous acceleration of the satellite is

(A) $-\frac{\alpha v^2}{2M}$ (B) $-\frac{\alpha v^2}{M}$ (C) $-\alpha v^2$ (D) $-\frac{2\alpha v^2}{M}$

Open solution

Q127. AIEEE 2012 (Online) 19th May

Sand is being dropped on a conveyer belt at the rate of $2\ \mathrm{kg}$ per second. The force necessary to keep the belt moving with a constant speed of $3\ \mathrm{m\,s^{-1}}$ will be

(A) $12\ \mathrm{N}$ (B) $6\ \mathrm{N}$ (C) zero (D) $18\ \mathrm{N}$

Open solution

Q128. AIEEE 2012 (Online) 19th May

A moving particle of mass m, makes a head on elastic collision with another particle of mass $2m$, which is initially at rest. The percentage loss in energy of the colliding particle on collision, is close to

(A) 33% (B) 67% (C) 90% (D) 10%

Open solution

Q129. AIEEE 2012 (Online) 12th May

A projectile moving vertically upwards with a velocity of $200\ \mathrm{m\,s^{-1}}$ breaks into two equal parts at a height of $490\ \mathrm{m}$. One part starts moving vertically upwards with a velocity of $400\ \mathrm{m\,s^{-1}}$. How much time it will take, after the break up with the other part to hit the ground?

(A) $2\sqrt{10}\ \mathrm{s}$ (B) $5\ \mathrm{s}$ (C) $10\ \mathrm{s}$ (D) $\sqrt{10}\ \mathrm{s}$

Open solution

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Q130. AIEEE 2012 (Online) 12th May

Open solution

Two bodies A and B of mass m and $2m$ respectively are placed on a smooth floor. They are connected by a spring of negligible mass. A third body C of mass m is placed on the floor. The body C moves with a velocity v_0 along the line joining A and B and collides elastically with A. At a certain time after the collision it is found that the instantaneous velocities of A and B are same and the compression of the spring is x_0. The spring constant k will be

(A) $m\frac{v_0^2}{x_0^2}$

(B) $m\frac{v_0}{2x_0}$

(C) $2m\frac{v_0}{x_0}$

(D) $\frac{2}{3}m\left(\frac{v_0}{x_0}\right)^2$

Q131. AIEEE 2012 (Online) 12th May

Open solution

A spring is compressed between two blocks of masses m_1 and m_2 placed on a horizontal frictionless surface as shown in the figure. When the blocks are released, they have initial velocity of v_1 and v_2 as shown. The blocks travel distances x_1 and x_2 respectively before coming to rest. The ratio $\left(\frac{x_1}{x_2}\right)$ is



(A) $\frac{m_2}{m_1}$

(B) $\frac{m_1}{m_2}$

(C) $\sqrt{\frac{m_2}{m_1}}$

(D) $\sqrt{\frac{m_1}{m_2}}$

Q132. AIEEE 2010

Open solution

Statement - 1 : Two particles moving in the same direction do not lose all their energy in a completely inelastic collision.

Statement - 2 : Principle of conservation of momentum holds true for all kinds of collisions.

(A) Statement - 1 is true, Statement - 2 is true; Statement - 2 is the correct explanation of Statement - 1

(B) Statement - 1 is true, Statement - 2 is true; Statement - 2 is **not** the correct explanation of Statement - 1

(C) Statement - 1 is false, Statement - 2 is true

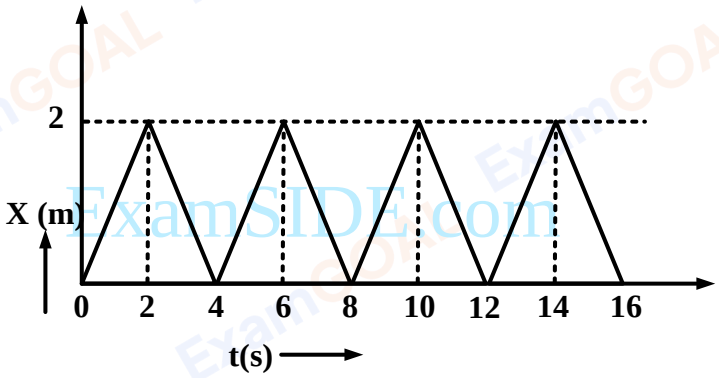
(D) Statement - 1 is true, Statement - 2 is false

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Q133. AIEEE 2010

The figure shows the position–time $(x-t)$ graph of one-dimensional motion of body of mass 0.4 kg. The magnitude of each impulse is

Open solution



(A) $0.4\ Ns$ (B) $0.8\ Ns$ (C) $1.6\ Ns$ (D) $0.2\ Ns$

Q134. AIEEE 2008

A block of mass $0.50\ kg$ is moving with a speed of $2.00\ ms^{-1}$ on a smooth surface. It strike another mass of $1.0\ kg$ and then they move together as a single body. The energy loss during the collision is :

Open solution

(A) $0.16J$ (B) $1.00J$ (C) $0.67J$ (D) $0.34\ J$

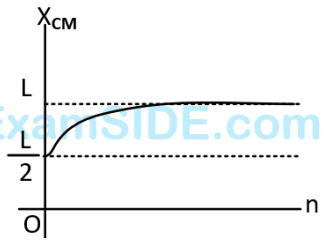
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Q135. AIEEE 2008

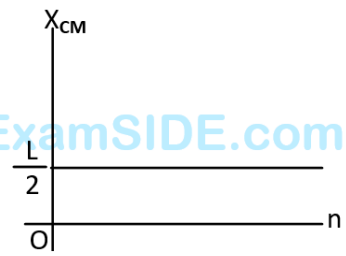
Open solution

A thin rod of length $'L'$ is lying along the x-axis with its ends at $x = 0$ and $x = L$. Its linear density (mass/length) varies with x as $k\left(\frac{x}{L}\right)^n$, where n can be zero or any positive number. If the position X_{CM} of the center of mass of the rod is plotted against $'n'$, which of the following graphs best approximates the dependence of X_{CM} on n?

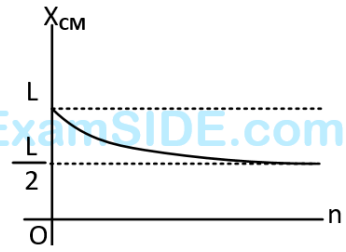
(A)



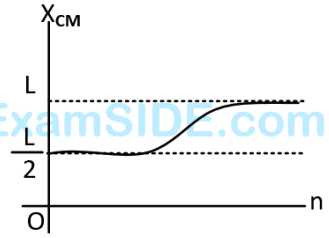
(B)



(C)



(D)



Q136. AIEEE 2007

Open solution

A circular disc of radius R is removed from a bigger circular disc of radius $2R$ such that the circumferences of the discs coincide. The center of mass of the new disc is αR from the center of the bigger disc. The value of α is

(A) $1/4$ (B) $1/3$ (C) $1/2$ (D) $1/6$

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Q137. AIEEE 2006

A player caught a cricket ball of mass $150\ g$ moving at a rate of $20\ m/s$. If the catching process is completed in $0.1s$, the force of the blow exerted by the ball on the hand of the player is equal to

(A) $150\ N$ (B) $3\ N$ (C) $30\ N$ (D) $300\ N$

Open solution

Q138. AIEEE 2006

A bomb of mass $16kg$ at rest explodes into two pieces of masses $4\ kg$ and $12\ kg$. The velocity of the $12\ kg$ mass is $4\ ms^{-1}$. The kinetic energy of the other mass is

(A) $144\ J$ (B) $288\ J$ (C) $192\ J$ (D) $96\ J$

Open solution

Q139. AIEEE 2006

Consider a two particle system with particles having masses m_1 and m_2. If the first particle is pushed towards the center of mass through a distance d, by what distance should the second particle is moved, so as to keep the center of mass at the same position?

(A) $\frac{m_2}{m_1}\ d$ (B) $\frac{m_1}{m_1+m_2}d$ (C) $\frac{m_1}{m_2}d$ (D) d

Open solution

Q140. AIEEE 2005

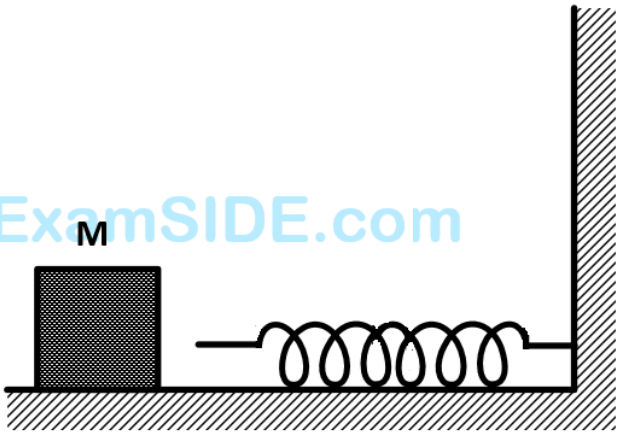
A body A of mass M while falling vertically downwards under gravity breaks into two-parts; a body B of mass $\frac{1}{3}\ M$ and a body C of mass $\frac{2}{3}\ M$. The center of mass of bodies B and C taken together shifts compared to that of body A towards

(A) does not shift (B) depends on height of breaking
(C) body B (D) body C

Open solution

Q141. AIEEE 2005

The block of mass M moving on the frictionless horizontal surface collides with the spring of spring constant k and compresses it by length L. The maximum momentum of the block after collision is



(A) $\frac{kL^2}{2M}$ (B) $\sqrt{Mk}\ L$ (C) $\frac{ML^2}{k}$ (D) Zero

Open solution

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Q142. AIEEE 2005

A mass $'m'$ moves with a velocity $'v'$ and collides inelastically with another identical mass. After collision the 1st mass moves with velocity $\frac{v}{\sqrt{3}}$ in a direction perpendicular to the initial direction of motion. Find the speed of the 2nd mass after collision.

Open solution

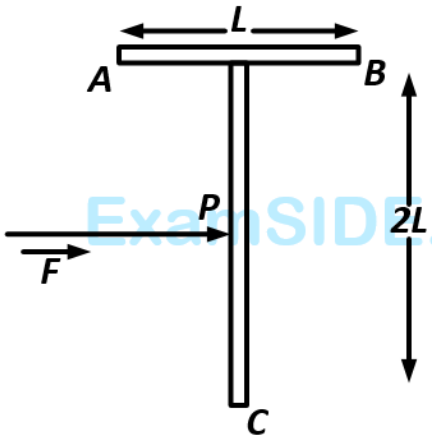


(A) $\sqrt{3}v$ (B) v (C) $\frac{v}{\sqrt{3}}$ (D) $\frac{2}{\sqrt{3}}v$

Q143. AIEEE 2005

A T shaped object with dimensions shown in the figure, is lying on a smooth floor. A force $'\vec{F}'$ is applied at the point P parallel to AB, such that the object has only the translational motion without rotation. Find the location of P with respect to C.

Open solution



(A) $\frac{3}{2}L$ (B) $\frac{2}{3}L$ (C) L (D) $\frac{4}{3}L$

Q144. AIEEE 2004

A machine gun fires a bullet of mass $40\ g$ with a velocity $1200ms^{-1}$. The man holding it can exert a maximum force of $144\ N$ on the gun. How many bullets can he fire per second at the most?

Open solution

(A) Two (B) Four (C) One (D) Three

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Q145. AIEEE 2003

Consider the following two statements :

A. Linear momentum of a system of particles is zero

B. Kinetic energy of a system of particles is zero.

then

(A) A does not imply B and B does not imply A

(B) A implies B but B does not imply A

(C) A does not imply B but B implies A

(D) A implies B and B implies A

Open solution

Q146. AIEEE 2002

Two identical particles move towards each other with velocity $2v$ and v respectively. The velocity of center of mass is

(A) v (B) $v/3$ (C) $v/2$ (D) zero

Open solution

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Answer Key

Numerical

1. **9**	2. **15**	3. **3**	4. **3**	5. **2**
6. **125**	7. **5**	8. **4**	9. **8**	10. **8**
11. **2**	12. **6**	13. **12**	14. **400**	15. **25**
16. **2**	17. **4**	18. **5**	19. **20**	20. **4**
21. **30**	22. **1**	23. **2**	24. **1**	25. **3**
26. **120**	27. **23**	28. **10**	29. **1**	

MCQ (Single Correct Answer)

30. **C**	31. **A**	32. **C**	33. **C**	34. **B**
35. **A**	36. **D**	37. **A**	38. **B**	39. **D**
40. **C**	41. **B**	42. **B**	43. **D**	44. **A**
45. **B**	46. **B**	47. **C**	48. **D**	49. **C**
50. **B**	51. **B**	52. **A**	53. **D**	54. **D**
55. **D**	56. **B**	57. **C**	58. **A**	59. **B**
60. **B**	61. **B**	62. **B**	63. **B**	64. **B**
65. **C**	66. **C**	67. **A**	68. **C**	69. **D**
70. **D**	71. **B**	72. **C**	73. **A**	74. **B**
75. **D**	76. **A**	77. **C**	78. **C**	79. **A**
80. **C**	81. **D**	82. **B**	83. **A**	84. **C**
85. **C**	86. **B**	87. **B**	88. **D**	89. **C**
90. **A**	91. **B**	92. **C**	93. **C**	94. **B**
95. **A**	96. **D**	97. **A**	98. **D**	99. **C**
100. **C**	101. **B**	102. **B**	103. **C**	104. **C**
105. **B**	106. **B**	107. **B**	108. **C**	109. **C**
110. **B**	111. **B**	112. **A**	113. **D**	114. **B**
115. **B**	116. **C**	117. *Scan QR*	118. **A**	119. **D**
120. **D**	121. **B**	122. **D**	123. **D**	124. **C**
125. **D**	126. **B**	127. **B**	128. **C**	129. **C**
130. **D**	131. **A**	132. **A**	133. **B**	134. **C**
135. **A**	136. **B**	137. **C**	138. **B**	139. **C**
140. **A**	141. **B**	142. **D**	143. **D**	144. **D**

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145. **A**	146. **C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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